

■ A.3.3 Modular Arithmetic

In doing mathematics, you often need to introduce objects that have special properties. This section gives an example of a package that deals with integers modulo n . The package works with *Mathematica* objects of the form `mod[a, n]`. It specifies the behavior of these objects under arithmetic operations such as addition and multiplication.

Notice the specification of transformation rules associated with the object `mod` for common operations such as addition. These rules essentially “overload” the standard addition function, to include special cases for sums involving the object `mod`.

```
Format[mod[a_,n_]] := SequenceForm[a," mod ",n]

mod[0,n_] := 0

mod[a_Integer,n_Integer] := mod[Mod[a,n],n] /; a>=n
mod[a_Integer,n_Integer] := mod[Mod[a,n],n] /; a<0

mod[b_Rational,n_Integer] :=
  mod[Mod[Numerator[b]*PowerMod[Denominator[b],-1,n],n],n]

mod /: mod[a_Integer,n_Integer] + b_Integer :=
  mod[Mod[a+b,n],n]
mod /: mod[a_Integer,n_Integer] + b_Rational :=
  mod[Mod[a + Numerator[b]*PowerMod[Denominator[b],-1,n],n],n]
mod /: mod[a_Integer,n_Integer] + mod[b_Integer,n_Integer] :=
  mod[Mod[a+b,n],n]
mod /: mod[a_Integer,m_Integer] + mod[b_Integer,n_Integer] :=
  Block[{g=GCD[m,n]}, mod[a,g]+mod[b,g] ]

mod /: mod[a_Integer,n_Integer] * b_Integer :=
  mod[Mod[a*b,n],n]
mod /: mod[a_Integer,n_Integer] * b_Rational :=
  mod[Mod[a*Numerator[b]*PowerMod[Denominator[b],-1,n],n],n]
mod /: mod[a_Integer,n_Integer] * mod[b_Integer,n_Integer] :=
  mod[Mod[a*b,n],n]
mod /: mod[a_Integer,m_Integer] * mod[b_Integer,n_Integer] :=
  Block[{g=GCD[m,n]}, mod[a,g]*mod[b,g] ]

mod /: mod[a_Integer,n_Integer] ^ q_Integer :=
  mod[PowerMod[a,q,n],n]
```

Modular arithmetic.

Read in the package for doing modular arithmetic.

```
In[1]:= <<ModularArithmetic.m
```

The package sets up a special print form for `mod` objects.

```
In[2]:= mod[2, 7]  
Out[2]= 2 mod 7
```

You can use standard arithmetic operators on `mod` objects; the definitions in the package specify that appropriate modular arithmetic operations get done.

```
In[3]:= %^19 + 6  
Out[3]= 1 mod 7
```